

# CMEP Project: Training a Conditional Variational Autoencoder (CVAE) on 2D Ising Model Data

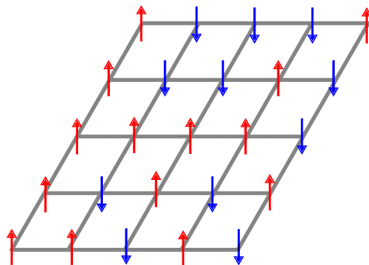
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# Ising model

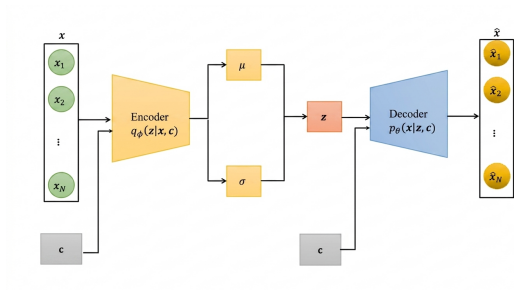
- ▶ Ising model data generated with Wolff cluster algorithm.
- ▶ Lattice dimension: 32x32.
- ▶ 135,000 samples for 45 temperatures (3,000 per temperature).
- ▶ Binary spin representation: up = 1, down = 0.



# Data Pipeline

- ▶ Training dataset of  $135,000 \times 32 \times 32$  for lattice samples
- ▶ Lattice dimension:  $32 \times 32$
- ▶ Condition dataset of 135,000 betas (inverse temperature)
- ▶ Normalize the beta conditions
- ▶ Datasets mixed and split using stratified sampling, split ratio = 0.8
- ▶ Load data into buffer and batch, batchsize = 128
- ▶ Apply on-the-fly data type conversion: for spins int8 to float32, betas float64 to float32
- ▶ (Optional) apply on-the-fly data augmentation
- ▶ On-the-fly processing is performed during training in parallel by the CPU

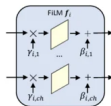
# CVAEs



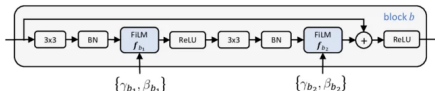
- Early embedding: tile beta condition and concatenate to the lattice channel
- Late embedding: concatenate beta condition to the latent code

# Feature-wise Linear Modulation (FiLM)

- ▶ This is a general-purpose conditioning layer in machine learning; it allows one layer to control or influence the internal computations of another layer by dynamically adjusting its data representations during training
- ▶ Dynamic Affine Transformation: Instead of using fixed weights, a FiLM layer applies an affine transformation (scaling and shifting) to the intermediate activations of a network:  $FiLM(condition)(x) = ax + b$
- ▶ The FiLM layer has a hidden layer (generator) to learn during training the affine transformations from the condition which are then applied in the output.
- ▶ Feature-Specific Modulation: The transformation is applied on a per-feature (or channel-wise) basis, allowing the network to promote or suppress information.
- ▶ The application here is to provide continuous late embedding to constantly remind the CNN not to forget the condition since it is [missing word/phrase].



(a) A FiLM layer.



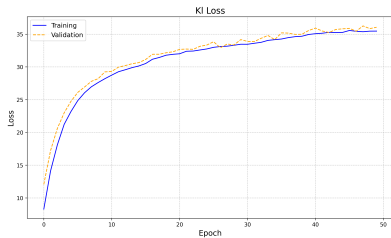
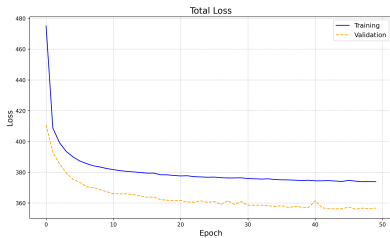
(b) A ResNet basic block with FiLM layers.

# Parameters and Hyperparameter tuning

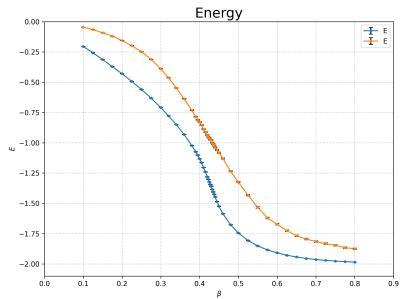
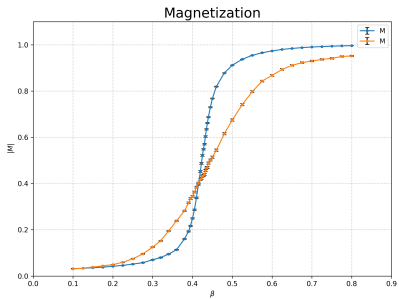
- ▶ Latent dimension: 64
- ▶ Encoder filters: [32, 64, 128, 256]
- ▶ Kernel: (3,3)
- ▶ strides: 2

| Parameter              | Search Space         | Best Value         |
|------------------------|----------------------|--------------------|
| Learning Rate          | $[10^{-6}, 10^{-3}]$ | $5 \times 10^{-5}$ |
| KL Weight ( $\alpha$ ) | [0.1, 5.0]           | 1.0                |

# Losses

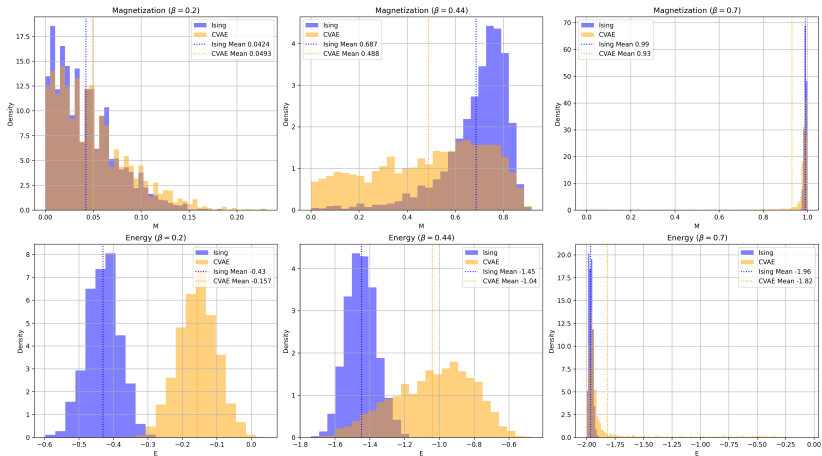


# Observables

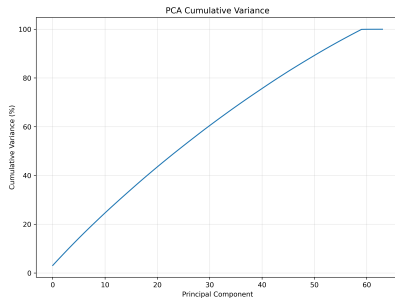




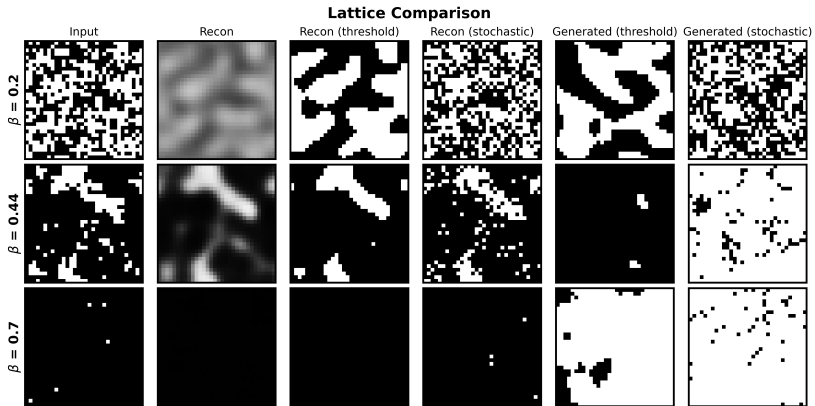
# Observable distributions



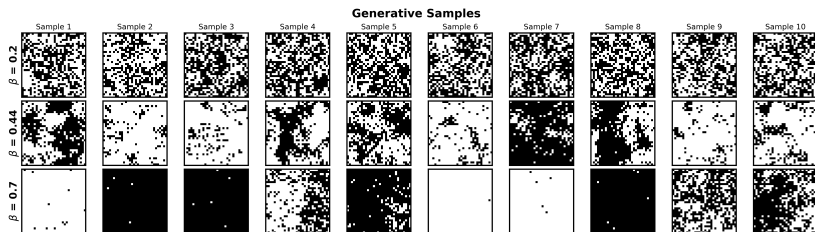
# Latent analysis



# Reconstructions I

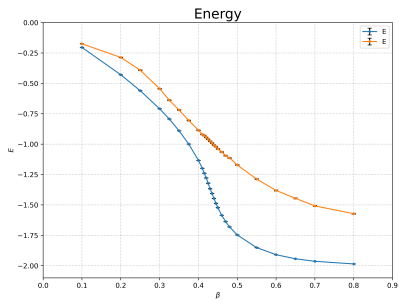
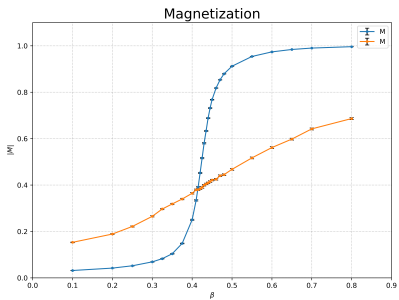


# Reconstructions II

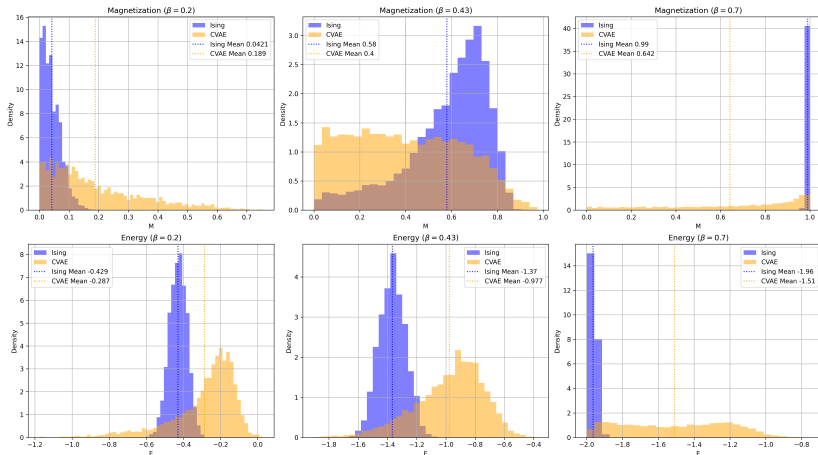


Thank You!

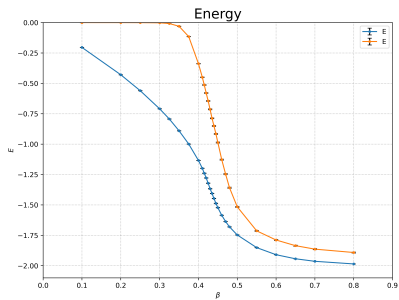
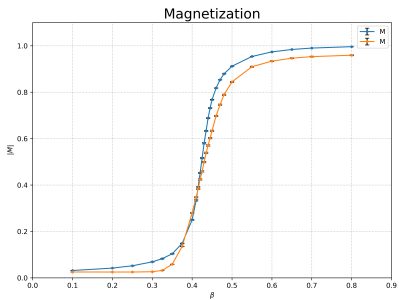
KL weight = 0.1



# KL weight = 0.1 distributions

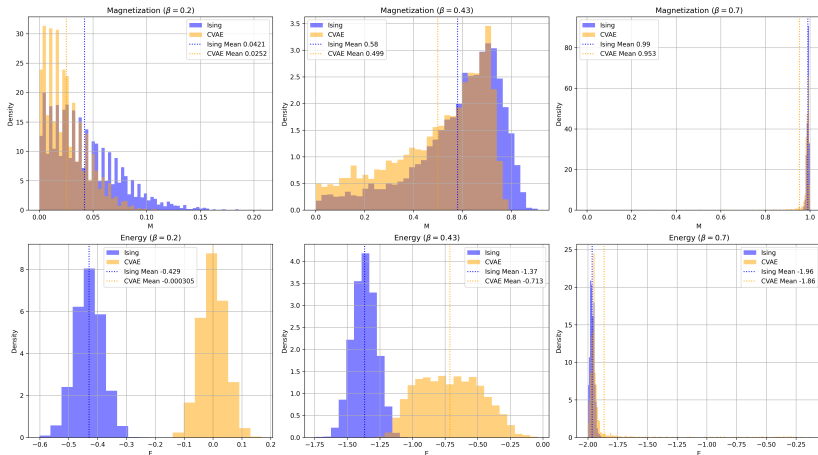


KL weight = 10.0





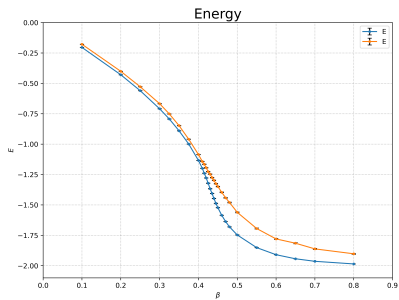
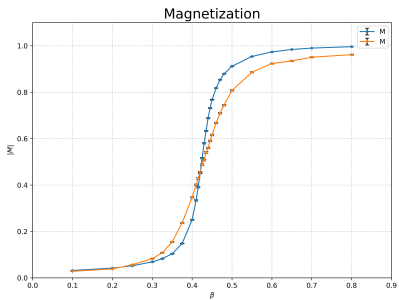
# KL weight = 10.0 distributions



# Physics losses

- ▶ Applied Gumbel-Softmax softmax annealing
- ▶ Total loss = Reconstruction loss + KL divergence + Magnetisation loss + Energy loss
- ▶ physics losses calculated with mean absolute error (MAE)

# Physics loss Observables



# Physics loss observables distributions

